Life as an Adaptive Self-organised Dissipative System

Cheong Jian Wei
School of Physical and Mathematical Sciences, Nanyang Technological University

The study of the nature of life has always been in the domain of biology. But if life was to emerge entirely via the laws of nature, we would expect to be able to describe the emergence, properties, and behaviours of life with a physical theory. A hint can be found in the stark contrast that differentiates life and other non-living inanimate matters, that is life's ability to efficiently capture or consume energy from its environment, and dissipating it as heat, in the process lowering its internal entropy while increasing the entropy of the environment that it is in. Such behaviour is one of a dissipative system in an open far from equilibrium system, driven by an external energy source such as the sun. In recent time, this description of life was met with rigorous thermodynamic definitions in what was referred to as dissipation-driven adaptation, capable of describing the emergence of life, its behaviour of self-replication, and its adaptation. In this letter, we look at recent and past developments in this emerging field, and the various *in vitro* and *in silico* studies that might be in support of this description of life.

I. INTRODUCTION

A. Background

The phenomenon of life has been a widely discussed topic in literatures since ancient times, with it being the focus of an entire modern field of biology. Biology study, describe, and characterise all aspects of life and its mechanisms. Modern biology was sparked by the widespread use of early microscopes in the 19th century, which eventually lead to the development of cell theory by Schwann and Schleiden, one of the core theory fundamental to biology. The theory of evolution arises in the same century with the Darwinian theory of natural selection, and goes hand in hand with the study of hereditary genetics.

The field of biology have made countless progress in these sub-fields, and our modern understanding of biology has given us a big picture on the phenomenon of life: Organic compounds formed in the early Earth around 4 billion years ago, which lead to the first life and eventually the last universal common ancestor (LUCA). All three domains of life, their kingdoms, and genera evolved from LUCA via darwinian evolution, with genetic diversity increasing the complexity of life generations by generations, eventually leading to complex life forms like us that are made of trillions of cells.

This put the focus on three main topics for discussion, the emergence of organic compounds and the first life known as abiogenesis, the mechanism behind self-replication that populates the Earth with life, and darwinian evolution bringing us from these simple organic compounds to the complex life forms that we know today. While darwinian evolution needs no explanation for the development of complex life other than mutation and exchange of genetic diversity, and natural selection or selective breeding, abiogenesis and how the first life self-replicate was a heavily debated topic.

Modern hypothesis of abiogenesis agreed on a RNA world before DNA and proteins are formed. The RNA world hypothesis was proposed by Alexender Rich in 1962 as a result of answering the chicken or egg paradox of whether DNA comes before or after proteins as one would require the other for self-replication [1]. RNA are able to function both as information carrier like DNA, and enzymes for synthesis, solving the chicken or egg problem, and allowing it to self-replicate. It should also be noted that the existence of a pre-RNA nucleotide before RNA is also a possibility [2].

The RNA world hypothesis answered the *what*, but not the *how*. How did organic compounds and complex structures like RNA formed from non-organic molecules and random structures in the first place? This brings us to the famed Miller-Urey experiment [3]. First suggested by Alexander Oparin, then J. B. S. Haldane, and thus also known as the Oparin-Haldane hypothesis; the primordial soup hypothesis claimed that organic compounds formed from non-organic ones simply because the early Earth's conditions favoured the synthesis of such organic compounds [4].

The Miller-Urey experiment simulated these early Earth's conditions in a lab setting. Starting from non-organic compounds such as methane, water, ammonia, and hydrogen, at least five amino acid monomers were synthesised after a week in the presence of a heat source, and simulated lightnings by electric sparks from a pair of electrodes. Thus, supporting the Oparin-Haldane hypothesis. Further analysis on Miller's other experiments such as the volcanic spark discharge experiment, and hydrogen sulfide spark discharge experiment showed that they synthesised more amino acid monomers than reported by Miller [5, 6], gaining further support for Oparin-Haldane hypothesis.

Once organic monomers and polymers which are

the basic building blocks of life starts to form and self-replicate, the formation of the first cell and eventually multicellular life can developed via adaptation to the environment, or mutation and exchange of genetic diversity [7]. However, the mechanisms behind the formation of the first cell from organic monomers and polymers are still not well understood. Similarly for other stages of life's evolution, its structures and complexities seems to make fast and abrupt transitions [8, 9], with increasing complexity.

B. Physical Descriptions

There are attempts to describe such irregularities that are not predicted by darwinian theories using statistical mechanics of phase transitions and self-organised criticality [10–12]. Such physical approaches to biology are becoming more and more widespread. For example, the increase in complexity with evolution has been described information-theoretically by showing that genomes behave like Maxwell's Demon in their environment [13]. Such physical theories, while not intended as a replacement for more in-depth biological descriptions, are promising in uniting the various disjointed biological theories and descriptions. By looking for a physical description of life, we are able to shine light on the possibility that the three main topics of focus that were described by different theories—abiogenesis, self-replication, and evolution, might arise from a common physical theory.

Physical interpretations of life are not a new idea. In Erwin Schrodinger's 1944 book—*What is Life?* [14], Schrodinger described life as a process that is feeding on "negative entropy", reducing its internal entropy and ejecting entropy to the surrounding environments. This was based on the observation that living things consume more ordered form of energies such as chemical potential energies in food, or UV wavelength light by chlorophyll, and emitting a more disordered form of energy—heat. This process ensures that the living organism stay alive, maintaining its low internal entropy, but dissipating entropy to the surroundings at the same time. Thus, this "negative entropy" is better described as nonequilibrium free energy that are either stored in, or obtained from the primary energy source that a living organism consume.

While Schrodinger's descriptions are able to explain how life does not violate the Second Law of thermodynamics despite seemingly so, a thermodynamic description of life was still missing. This is because the framework for open systems far from equilibrium, such as life, were still not well defined. Progress in weakly driven nonequilibrium systems were made in the 1960s by Ilya Prigogine [15]. Prigogine coined the term dissipative structures to describe dynamical steady state structures that appear in these far from equilibrium systems, winning him the Nobel Prize in Chemistry in 1977 [16].

Some examples of dissipative structures are vortices in turbulent flows, their atmospheric counterparts—tornadoes and hurricanes, and convection cells. These are structures that maintain their stable structures while dissipating energy to the surroundings, similar to Schrodinger's description of life. Prigogine himself has described biological systems as a dissipative system, and has emphasised irreversibility and the arrow of time by using the aging process of biological system as an example [17].

Since then, multiple works have surfaced to describe life as a dissipative system. For example, Schneider and Kay described biological evolution as responses to dissipating gradients, and ecosystems as energy degraders, working together to degrade energy gradients efficiently [18]. Descriptions of life's emergence as self-organisation at the edge of chaos has also been shown, whereby complexity is maximum and energy consumption is neither too high or too low [19, 20]. Karo Michaelian from the National Autonomous University of Mexico has also applied nonequilibrium thermodynamics directly to ecosystems to show its thermodynamic stability [21], and demonstrated that a dissipative system description is consistent with the formation and self-replication of RNA polymers in the early Earth [22].

Such nonequilibrium descriptions of life was brought to attention again when Jeremy England from the Massachusetts Institute of Technology derived a generalised Second Law of thermodynamics for dissipative systems, allowing the behaviours of systems such as life, to be predicted in the presence of a strongly driven external energy source [23]. This is referred to as dissipation-driven adaptation or dissipation-driven adaptive organisation.

II. DISSIPATION-DRIVEN ADAPTATION

While in an equilibrium system, configurations obey Boltzmann distribution. This means that the ratio of the probability of transition from configuration i to j or k is proportional to the exponential of the energy difference of the two configurations.

$$\frac{\Pr(i \to j)}{\Pr(i \to k)} = \exp\left[\frac{E_k - E_j}{k_B T}\right] \tag{1}$$

This means that at thermal equilibrium, the behaviours of the equilibrium system can be predicted based on the energy of the system alone. However, in a presence of an external drive, the Boltzmann distribution no longer holds. The heat dissipation now depends on

both the driven work and the change in internal energy (equation 3), whereas in the equilibrium case, the heat dissipation is simply the change in internal energy due to conservation of energy (equation 2).

$$\Delta Q_{\text{equilibrium}} = -\Delta E_{i\to j} \tag{2}$$
$$\Delta Q_{\text{driven}} = W_{i\to j} - \Delta E_{i\to j} \tag{3}$$

This complication is where Prigogine's formulation which builds on Onsager's formulation of reciprocal relations in nonequilibrium systems with local equilibrium [24] is useful. However, Prigogine's formulation ultimately breaks down when used to describe systems such as life, that cannot be described based on local equilibriums [25]. This problem of nonequilibrium systems was finally resolved when Jarzynski and Crooks derived the fluctuation theorem [26–28].

$$\frac{\Pr(\Delta S)}{\Pr(-\Delta S)} = \exp[\Delta S] \tag{4}$$

The fluctuation theorem as described in equation 4 holds even in systems with a strong external drive. At risk of oversimplification, equation 4 simply tells us that the Second Law is only true on average—transitions can result in a negative change in entropy, but this is exponentially rare. This imply that Second Law still holds, but only on average, i.e. $\langle \Delta S \rangle \geq 0$ where the equality holds if and only if $\Delta S = 0$.

Using this relation, England derived a generalised Second Law as shown in equation 5 [23].

$$\beta \langle \Delta Q \rangle_{X\to X'} + \ln\left[\frac{\pi(X' \to X)}{\pi(X \to X')}\right] + \Delta S_{int} \geq 0 \tag{5}$$

Where we have simply let $\pi(X \to X')$ be the probability of transiting from configuration X to X', and expressed the total change in entropy ΔS of the transition $X \to X'$ to be $\beta\langle \Delta Q \rangle_{X\to X'} + \Delta S_{int}$, i.e. the sum of the entropy dissipated to the bath and the change in internal entropy. Equation 5 is simply the familiar Second Law of $\Delta S_{tot} \geq 0$ with the addition of an extra term describing the irreversibility of the transition.

An irreversible process gives $\pi(X' \to X) < \pi(X \to X')$. Hence, the more irreversible a process is, the more negative the term $\ln\left[\frac{\pi(X'\to X)}{\pi(X\to X')}\right]$, implying a higher entropy production. Additionally, a very probable transition, i.e. $\pi(X \to X')$ is large and close to 1, almost ensure that the irreversibility term is negative. Therefore, one could interpret this as: it is more probable for the system to transit to a state whereby entropy production to the environment is positive. With this, we have captured the first consequence of England's theory of dissipative adaptation, that dissipative systems naturally favoured configurations that dissipates entropy to the environment.

The generalised Second Law can further describe the self-replication process. For population $n(t)$ at time t, if we were to assume exponential growth, then we have

$$n(t) = n(0)e^{(g-\delta)t} \tag{6}$$

Thus, if we were to describe the irreversible process of self-replication as $X \to X'$, we have the probability of self-replication in an interval dt as $\pi(X \to X') = g\ dt$. Whereas the probability of the reverse transition whereby a copy was to return to and merge with its parent, or if it were to die or decay, the probability would be $\pi(X' \to X) = \delta\ dt$. Therefore, the generalised Second Law becomes

$$\beta \langle \Delta Q \rangle_{X\to X'} + \ln\left[\frac{\delta}{g}\right] + \Delta S_{int} \geq 0 \tag{7}$$

Again, the more negative the term $\ln\left[\frac{\delta}{g}\right]$, the higher the entropy production must be for the generalised Second Law to hold. This brings us to the second consequence of dissipative adaptation, that self-replication and increasing the number of dissipators, i.e. $g > \delta$, and hence $\ln\left[\frac{\delta}{g}\right] < 0$, increases the entropy production. Note that while exponential growth was used in the example, the principle also applies for other models of population growth.

The generalised Second Law only describes the probability of transiting from an initial state to a final state. However, in general there are multiple final states X', X'', X''' etc. and it is natural for us to ask the question of which transition is more likely, i.e. comparing the probabilities of transiting to two distinct states, $\pi(X \to X')$ and $\pi(X \to X'')$. England's group has derived a relation for the ratio of the two probabilities which can be regarded as a generalised free energy [29].

$$\ln\left[\frac{\pi(X \to X')}{\pi(X \to X'')}\right] = -\Delta \ln\left\langle \frac{p_f}{p_{bz}} \right\rangle_{X',X''} + \ln\left[\frac{\pi(X' \to X)}{\pi(X'' \to X)}\right] + \Delta\Psi_{X',X''} - \Delta\Phi_{X',X''} \tag{8}$$

Whereby p_f and p_{bz} is the final probability distribution and the Boltzmann distribution respectively, and $\Delta\Psi_{X',X''} - \Delta\Phi_{X',X''}$ is the difference in average dissipated work Ψ between X' and X'', a positive value

implies that transition to state X' dissipates more work than X''. The term Φ is simply a correction term due to the dispersion of the distribution. In other words, the cumulant generating function gives

$$-\ln\langle\exp(-\beta W_d)\rangle = \beta\langle W_d\rangle - \frac{\beta^2}{2}\sigma^2_{W_d} + \dots \equiv \Psi - \Phi \tag{9}$$

The details of the derivation of equation 8 will not be shown here, but in short, it is simply the ratio of the fluctuation theorem equations for the two transitions $X \to X'$ and $X \to X''$, and the p_f/p_{bz} term comes from an expression of relative entropy.

Equation 8 tells us that the ratio of the probabilities of transiting to X' or X'' is proportional to the difference in work dissipation $\Delta\Psi$ between X' and X'', in other words, if X' were to absorb and dissipate more work, the probability of transiting to that state X' will be higher. The term of p_f/p_{bz} is zero if the final probability distribution p_f is the equilibrium Boltzmann probability p_{bz}, this means that transitions to equilibrium states are also more probable. The final term of $\pi(X' \to X)/\pi(X'' \to X)$ simply says that a more reversible transition is also more probable. This completes England's dissipative adaptation with the last consequence, that amidst transitions to equilibrium and reversible states, transitions also favours final states that dissipates the most heat.

The three consequences describe the behaviours that a system acting according to dissipation-driven adaptation must exhibits. Bringing the focus back to biological life, this means that if life is a dissipative system as a consequence of England's theory of dissipative adaptation, it should: 1) self-organised to structures or configurations that dissipate entropy to the environment efficiently, 2) self-replicates to dissipate more entropy, and lastly 3) its adaptation to changing environments or evolution are transitions to new states or configurations that efficiently disipates entropy in that environment. For the rest of this letter, we will review some studies that might be in support of this hypothesis of life as a dissipation-driven adaptive self-organised system.

A. Self-organisation

Self-organisation of complex structures in the presence of an external energy source have been demonstrated both experimentally and computationally. England's group have conducted a computational simulation of twenty particles in a toy chemical space, and undergoing periodic driving [30]. The system of twenty particles interacts with each other via a bonding energy that breaks or forms based on distance between particles. By driving a single particle with a fixed periodic force, the system's normal mode peaks at the driving frequency, exhibiting resonance. Additionally, the rate of work absorption and heat dissipation increases. Whereas, in a system without the periodic driving, the particles eventually cluster together randomly.

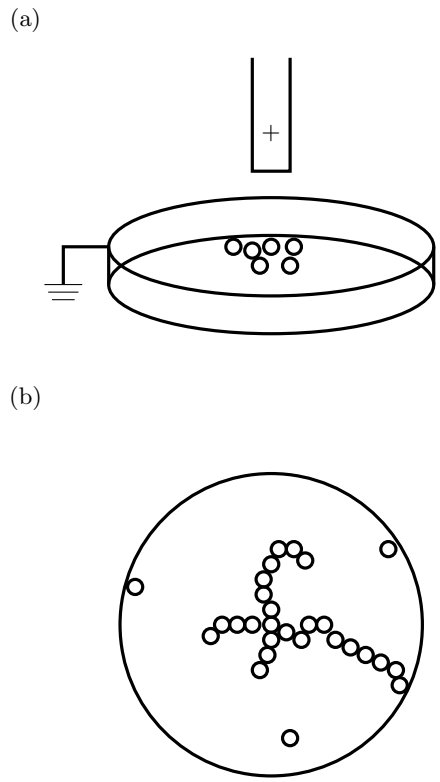

FIG. 1. Experiment conducted by Kondepundi et al. [31]. (a) Conducting beads in oil with a voltage potential. Forty conducting beads were used in the experiment (only a few are shown in the diagram), the conducting beads' initial positions were at the centre of the petri dish, clustering together. (b) An example of self-organisation or self-assembly of complex structures forming after some time. The conducting beads form tree-like structures. A critical transition occurs when the conducting beads self-organised, a spike in entropy dissipation was also observed after formation of complex structure.

Such self-organisation of complex structures with an external force was also shown to occur in macroscopic systems. One very intriguing experiment was conducted by Kondepudi et al. [31]. In the experiment, conducting beads 4mm in diameter was immersed in oil in a petri dish. A high voltage of 15kV to 30kV was applied to the system via an electrode above the petri dish, together with a metal ring surrounding the oil and conducting beads serving as the ground electrode as shown in FIG. 1.

After a period of time, the conducting beads self-organised to a complex tree-like structure from its initial randomly clustered positions. This formation was not

gradual, a critical point was reached and the conducting beads self-organised almost immediately. Even more surprising, there is a spike in entropy dissipation right when the conducting beads' structure reached a steady state. The entropy dissipation was defined as:

$$\Sigma = d_i S/dt = VI(X_i, v(r))/T \tag{10}$$

Whereby V is the applied voltage, I is the current, which depends on the beads' positions X_i, and the velocity field of the oil $v(r)$, and T is the temperature.

An increase in the applied voltage increases the change in entropy dissipation $\Delta\Sigma$ before and after formation of complex structures, albeit only by a relatively small amount. On the other hand, an increase in the number of conducting beads resulted in a huge jump in $\Delta\Sigma$. The complex structure was also shown to be self-healing, that is when the structure is broken by the experimenter, the conducting beads will build the tree-like structures itself again in a few seconds.

The conducting beads experiment only varied the applied voltage and the number of beads, while showing that self-organisation indeed takes place in a presence of an external driving force and works to dissipate entropy efficiently, but it does not show how the system's self-organised complex structures might organise themselves to be suited for the external source.

For that, another experiment by Ito et al. [32] performed a similar experiment on silver nanorods, whereby the nanorods are shown to self-organised in the presence of a doughnut-shaped laser source. It was found that the silver nanorods self-organised into differing structures that depended on the wavelengths and polarisation of the laser source. Hence, this experiment demonstrated how the self-organised structures might varies with the energy content of the external source, which was not demonstrated in the conducting beads experiment.

All these experiments supports England's dissipative adaptation of life hypothesis. However, does this conform with what we know about abiogenesis? Did early organic compounds self-organised themselves because of dissipative adaptation with the sun or hydrothermal vents as the external nonequilibrium energy source? Michaelian's group have been answering these questions in recent years, by taking the external nonequilibrium source as UVC light of 230nm to 270nm from the Sun which was one of the ranges of wavelengths that were most intense in the early Earth [33].

Michaelian demonstrated how RNA polymers proliferate via photochemical reactions and how DNA polymers denature from absorption and dissipation at UVC wavelengths, which is a crucial step for self-replication without enzymes [34]. Homochirality was also shown to be achieved in DNA and RNA via photochemical reactions with UVC wavelengths light [35]. Interestingly, Michaelian brought into attention the fact that correlations exist between the maximum absorbance wavelengths of early life's nucleobases and amino acids, and UVC wavelenghts [36]. Some examples are shown in TABLE I. This is very strong evidence that in the RNA world during early Earth era, RNA or pre-RNA polymers self-organised or adapted to absorb and dissipate UVC solar spectrum efficiently, a feature of dissipative adaptation.

Molecules	Max absorbance (nm)
Adenine	261
Guanine	243, 272.75
Thymine	263.8
Cytosine	266.5
Uracil	258.3
Phenylalanine	257.5
Tyrosine	274.2
Tryptophan	278
Histidine	211
Thiamine	235, 267
Riboflavin	263, 346, 447

TABLE I. Nucleobases and amino acids that are the foundations of life. Their maximum absorbance wavelengths coincides roughly with UVC solar spectrum of 230nm to 270nm. This hints that these molecules self-organised or adapted to efficiently absorb and dissipate UVC solar spectrum, exhibiting dissipative adaptation. [36]

B. Self-replication

Quite intuitively, we would expect having more dissipators to increase the overall rate of dissipation. Hence, it should be quite clear that the behaviour of self-replication work towards increasing overall rate of dissipations, in accordance with the hypothesis of dissipative adaptation of life. This result was also shown in the conducting beads experiment whereby increasing the number of beads resulted in an increased spike in entropy dissipation. However, some care have to be taken in this assumption, as it might be possible for a system to dissipate more heat by itself as opposed to undergoing the self-replication process. This would be true if the self-replication process and the time in which it takes place produces lesser entropy than if the system was to dissipate heat alone faithfully, or if individual dissipators became less efficient in the presence of other dissipators.

In the same paper where England derived his gen-

eralised Second Law for self-replication [23], it was estimated that RNA's self-replication has a heat production rate greater or equal to 7 kcal mol^{-1}, which was close to the reaction's enthalpy of 10 kcal mol^{-1} obtained experimentally. It was also shown that a *E. coli* bacterium can dissipates at least 5 times as much heat as its individual heat dissipation when it self-replicates. This result was obtained from a another expression of equation 7

$$\beta\langle Q\rangle \geq 2n_{pep}(\tau_{div}/t_{\max}+1) - \Delta S_{int} \tag{11}$$

Whereby τ_{div} is the division time interval, $t_{\max}$ is the lifetime and n_{pep} is the number of peptides bond. The growth rate would be approximately n_{pep}/τ_{div} [37]. This gives a $\beta\langle Q\rangle$ of $220n_{pep}$ when self-replicating and $\beta\langle Q\rangle$ of $42n_{pep}$ when dissipating heat individually.

Some results have also shown that DNA copolymerisation, i.e. the process of replicating or copying DNA polymer, dissipates entropy that might be affected by the information entropy of the DNA polymer [38]. As DNA copolymerisation takes place in the presence of a driving force that is chemical in nature, information is created at the copied DNA polymer, decreasing the Shannon entropy of itself and generating mutual information with the original copy. However, in the process, entropy is dissipated to the surrounding. This is quantified by the affinity per copied nucleotide. As driving force increases, from equilibrium, there is an increase in entropy production that increases with the driving force.

In general, most dissipative systems exhibit behaviours of self-replications. For example, computer simulations of colloidal clusters show that the exponential growth rate of self-replication varies with the energy landscape of the system [39]. Three-dimensional vortices also self-replicates in rotating shear flows, as demonstrated in another numerical study [40], this was also proposed to be how planetoids form in protoplanetary disks [41], whereby dissipative system such as a protoplanetary disk can replicate smaller dissipative systems in it, or eventually splits into a copy or more of itself in phenomenon such as disk fragmentation [42] and forms binary star systems.

Thus, by itself, the phenomenon of self-replication in complex life and organic polymers and monomers were already a feature of dissipative system. With the addition of the self-replication process dissipating more entropy as demonstrated in *E. coli* replication and DNA copolymerisation, biological self-replication agrees with England's hypothesis of life as a dissipative system obeying dissipative adaptation. One might even suggest that early organic compounds evolved the ability self-replicate to ensure that the presence of such dissipative structures are constant.

C. Adaptation

If life does indeed self-organise to better absorb the surrounding external energy source and to dissipate heat, we would expect it to change or adapt its structures with changes in external energy sources. Throughout the history of life, life's evolution has underwent avalanche dynamics, whereby there are quick and sudden changes in complexity instead of a smooth and gradual evolution [12]. Some has even modelled this sudden transition of evolution as a Big Bang model with evolutionary inflation akin to the cosmological inflation model [9]. Self-organised criticality with fractal scaling and $1/f$ noise that were valid for various critical phenomena like directed percolation or interface depinning, were shown to also be valid for a specific model of biological evolution known as the Bak-Sneppen model [10, 43].

This was explicitly demonstrated in simulated artificial life [11], whereby the microscopic processes can be controlled and fine-tuned to simulate similar behaviours to biological life such as mutation and self-replication. The artificial life simulation is the Tierra system, whereby computer instructions are the simulated life that can self-replicate in the memory, and have random mutations and natural selection [44]. In the simulation, it was shown that the fitness or how the population adapts to the environment, underwent multiple discontinuous jumps, the height of the jumps can be both small and large, and the period of the jumps varies widely too.

Even if we were to be convinced that evolution exhibits self-organised criticality behaviours, for England's dissipative adaptation to hold, we requires these critical transitions to work towards absorbing more work and dissipating more heat. This result was shown again by Michaelian, similar to their results on the correlations between early life molecules' maximum absorbance and the UVC solar spectrum, it was shown in the same paper that there are also correlations between the change in solar spectrum throughout Earth's history and the emergence of more complex life forms [36].

Michaelian partitioned Earth's history into five different periods: 3.85 – 3.2 Ga, 3.2 – 2.7 Ga, 2.7 – 2.45 Ga, 2.45 – 2.2 Ga, and 2.2 Ga – present. Whereby Ga is giga-annum, and a Ga is a billion years ago. These periods were partitioned based on changes in the environment on the Earth's surface. Such conditions include: solar spectrum, temperature, and atmospheric compositions. Complex organic compounds and cells evolved in different periods of time to better survive or absorb energy according to the environmental conditions in that period. A summary of the correlations is shown

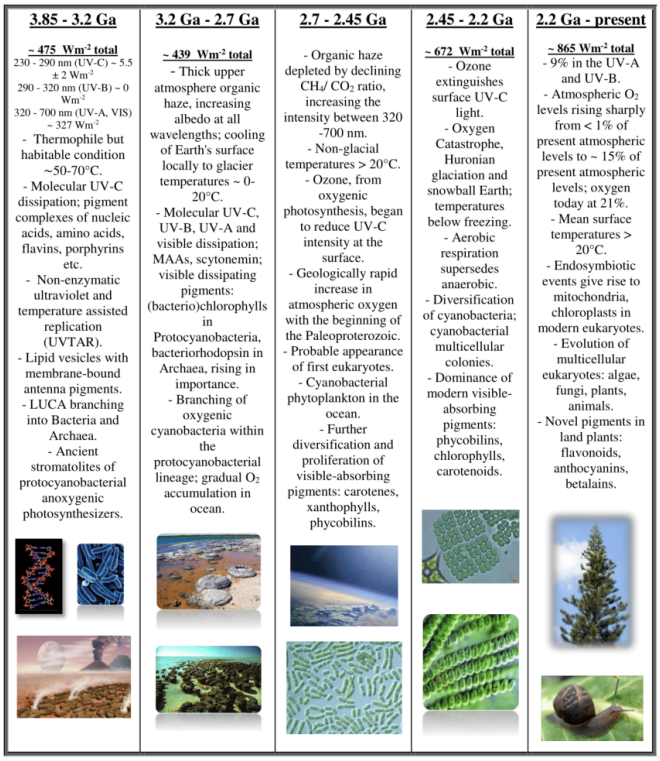

FIG. 2. Correlations between solar spectrum and the emergence of organic compounds and cells in the five different periods. As Earth's conditions changes, more complex organic compounds evolved to suit these conditions. The general trends are a shift from UVC absorbing to visible spectrum absorbing compounds, and a shift to oxygen based respiration. Reproduced from Ref. [36] licensed under CC BY 3.0.

in FIG. 2.

With addition to evolving or adapting to self-organised structures that are efficient energy absorbers and dissipators, the processes sustaining biological life can also be optimised for dissipation. One specific example is the catalytic dwell of ATP or adenosine triphosphate synthase motor. ATP is the energy unit of cells, an ATP synthase is an enzyme that lies along the cell membrane of a cell, and will break ATP down to ADP or adenosine diphosphate and a phosphate. This breakage of bond releases energy that powers the cell. This process is a rotary motion driven by an electrochemical potential, also know as the proton motive force, that is caused by a difference in proton concentrations. The ATP synthase will rotate in angles of 120°, for each ATP to ADP + P_i reaction. However, at the moment of breakage of ATP bonds to ADP + P_i, the synthase will pause for a short amount of time of around 2ms [45], the angle at which this pause happens is known as the catalytic dwell.

By using a kinetic model of ATP synthase, Dewar et al. [46] showed that both the entropy production during the ATP synthase process, and the Shannon entropy of the transitional states, peaks at a fixed catalytic dwell angle. Both peaks almost coincides with each other at relative angle of 0.598 and 0.602 rads respectively. This coincides with the actual catalytic dwell of 0.6 rad obtained by fitting the kinetic model to experimental data. Therefore, it is concluded that the ATP synthase's functionalities and specifically the angle of the catalytic dwell, has evolved to be consistent with maximum entropy production.

Michaelian's work in discerning the correlations between the emergence or evolution of organic compounds with the change in solar spectrum on Earth's surface, is indeed very strong evidence that the process of self-organsation to external energy source does not stop after initial self-organisation, but carried on to adapt to changing conditions, increasing complexity at each step. While this correlation might be a coincidence, the study on ATP synthase's catalytic dwell is a good example that biological processes does indeed are optimised for entropy dissipation, even if the reasons or mechanisms behind this unclear.

III. CONCLUSION AND FUTURE WORKS

England's dissipative-driven adaptation describes dissipative structures that emerges in an open far from equilibrium system. A few consequences of such systems are: first, the emergence or self-organisation of complex structures from random configurations, whereby these complex structures will be optimised for the absorption of the external energy source, a phenomenon which have been compared to the resonance effect by England's group [30]. Second, the ability to self-replicate to create smaller self-similar structures in its own structure, or a faithful copy of itself to further dissipate entropy. Lastly, this self-organisation or resonance behaviour is continuous, and the complex structure continues to adapt and self-organised to change in the external energy source to better dissipate heat. A physical interpretation of life find similarities in life's behaviours and these consequences, and thus a natural question is whether life is indeed simply a adaptive self-organised dissipative system.

If this were true, then a new perspective on life not possible in biological theories can be achieved, whereby the various biological processes, phenomena and behaviours could be explained by a single theory of dissipative adaptation. While we have only discussed self-organisation, self-replication, and adaptation, one might wonder what other biological processes and behaviours might obey such a principle of dissipative adaptation, and perhaps understanding of such processes might even bring about advances in the field of medicine. Additionally, such new perspective might loosen the constraints on "life as we know it" in our search for

extraterrestrial lifeforms, such that the conditions required for life might be not as tight as we expect, and life might more common and unfamiliar than we thought.

Evidences from various experimental and computational studies seems to be consistent with England's hypothesis that life is indeed such dissipative systems, with its purpose being to dissipate entropy or heat efficiently as with other dissipative systems. The phenomena of self-organisation, self-replication, and adaptation as a process for optimising entropy dissipations, were demonstrated most significantly, first by the conducting beads experiment for self-organisation, which showed that self-organisation of complex structures lead to an increase in entropy dissipation, second by the estimate of entropy produced in *E. coli* replication and a simulation of DNA copolymerisation for self-replication, which showed that the self-replicating process generates more entropy production, and lastly by ATP synthase modelling using kinetic models for adaptation, which showed that biological processes has also evolved to maximally dissipate entropy. Additionally, Michaelian's group have also made great contributions in support of the hypothesis by analysis of the maximum absorbance of individual organic compounds or "pigments" as it was referred to in their works. The emergence of these compounds seems to correlate with the environmental conditions by which they arosed in, in an effort to better survive or absorb and dissipate energy.

However, there are still unanswered questions and experimental confirmations are required for the hypothesis to be conclusive. For example, while correlations between early organic compounds and the solar spectrum are shown, an *in vitro* experiment to demonstrate if the early organic compounds indeed self-organised to the solar spectrum must be conducted, similar to the Miller-Urey experiment. Additionally, how does natural selection or sexual selection comes into play? A recent study on *Drosophilia* flies showed that in a controlled environment without the hurdles of natural selection, the complexity of the features of the flies increases over generations [47], which might suggest that natural selection actually hinders the development of complexity in evolution, i.e. species' genes favours simplicity if it works towards the species survival. Also, how does the emergence of two primary sexes arises based on the principle of dissipative adaptation, and if it does not, what other emergent properties of life that are not encompassed by this principle? Perhaps one might even wonder if life is a consequence of dissipative adaptation, or if dissipative adaptation is simply one of the many behaviours of life to ensure the survival of the species by optimising absorption of available energy. At this point of time however, the hypothesis of life as an adaptive self-organised dissipative system is promising, and many future works and investigations remain to be performed in this new and emerging field.

[1] M. Neveu, H.-J. Kim, and S. A. Benner, The "strong" RNA world hypothesis: Fifty years old, Astrobiology **13**, 391 (2013).

[2] T. R. Cech, The RNA worlds in context, Cold Spring Harbor Perspectives in Biology **4**, a006742 (2011).

[3] S. L. Miller, A production of amino acids under possible primitive earth conditions, Science **117**, 528 (1953).

[4] S. Tirard, J. b. s. haldane and the origin of life, Journal of Genetics **96**, 735 (2017).

[5] A. P. Johnson, H. J. Cleaves, J. P. Dworkin, D. P. Glavin, A. Lazcano, and J. L. Bada, The miller volcanic spark discharge experiment, Science **322**, 404 (2008).

[6] E. T. Parker, H. J. Cleaves, J. P. Dworkin, D. P. Glavin, M. Callahan, A. Aubrey, A. Lazcano, and J. L. Bada, Primordial synthesis of amines and amino acids in a 1958 miller h2s-rich spark discharge experiment, Proceedings of the National Academy of Sciences **108**, 5526 (2011).

[7] H. Follmann and C. Brownson, Darwin's warm little pond revisited: from molecules to the origin of life, Naturwissenschaften **96**, 1265 (2009).

[8] W. C. Ratcliff, R. F. Denison, M. Borrello, and M. Travisano, Experimental evolution of multicellularity, Proceedings of the National Academy of Sciences **109**, 1595 (2012).

[9] E. V. Koonin, The biological big bang model for the major transitions in evolution, Biology Direct **2**, 21 (2007).

[10] S. Maslov, M. Paczuski, and P. Bak, Avalanches and1fnoise in evolution and growth models, Physical Review Letters **73**, 2162 (1994).

[11] C. Adami, Self-organized criticality in living systems, Physics Letters A **203**, 29 (1995).

[12] M. Paczuski, S. Maslov, and P. Bak, Avalanche dynamics in evolution, growth, and depinning models, Physical Review E **53**, 414 (1996).

[13] C. Adami, C. Ofria, and T. C. Collier, Evolution of biological complexity, Proceedings of the National Academy of Sciences **97**, 4463 (2000).

[14] E. Schrodinger and R. Penrose, WHAT IS LIFE?, in *What is Life?* (Cambridge University Press) pp. ix–ix.

[15] I. Prigogine and J. Kestin, Introduction to thermodynamics of irreversible processes (second revised edition), Journal of Applied Mechanics **31**, 368 (1964).

[16] I. Prigogine, Time, structure, and fluctuations, Science **201**, 777 (1978).

[17] I. Prigogine, Irreversibility and space-time structure, in *Springer Proceedings in Physics* (Springer Berlin Heidelberg, 1984) pp. 2–10.

[18] E. Schneider and J. Kay, Life as a manifestation of the second law of thermodynamics, Mathematical and Computer Modelling **19**, 25 (1994).

[19] T. S. Ray and N. Jan, Anomalous approach to the self-organized critical state in a model for "life at the edge of chaos", Physical Review Letters **72**, 4045 (1994).

[20] P. T. Macklem, Emergent phenomena and the secrets of life, Journal of Applied Physiology **104**, 1844 (2008).

[21] K. Michaelian, Thermodynamic stability of ecosystems, Journal of Theoretical Biology **237**, 323 (2005).

[22] K. Michaelian, Thermodynamic dissipation theory for the origin of life, Earth System Dynamics **2**, 37 (2011).

[23] J. L. England, Statistical physics of self-replication, The Journal of Chemical Physics **139**, 121923 (2013).

[24] L. Onsager, Reciprocal relations in irreversible processes. i., Physical Review **37**, 405 (1931).

[25] I. Prigogine and G. Nicolis, Biological order, structure and instabilities, Quarterly Reviews of Biophysics **4**, 107 (1971).

[26] J. Kurchan, Fluctuation theorem for stochastic dynamics, Journal of Physics A: Mathematical and General **31**, 3719 (1998).

[27] G. E. Crooks, Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences, Physical Review E **60**, 2721 (1999).

[28] D. Collin, F. Ritort, C. Jarzynski, S. B. Smith, I. Tinoco, and C. Bustamante, Verification of the crooks fluctuation theorem and recovery of RNA folding free energies, Nature **437**, 231 (2005).

[29] N. Perunov, R. A. Marsland, and J. L. England, Statistical physics of adaptation, Physical Review X **6**, 10.1103/physrevx.6.021036 (2016).

[30] T. Kachman, J. A. Owen, and J. L. England, Self-organized resonance during search of a diverse chemical space, Physical Review Letters **119**, 10.1103/physrevlett.119.038001 (2017).

[31] D. Kondepudi, B. Kay, and J. Dixon, End-directed evolution and the emergence of energy-seeking behavior in a complex system, Physical Review E **91**, 10.1103/physreve.91.050902 (2015).

[32] S. Ito, H. Yamauchi, M. Tamura, S. Hidaka, H. Hattori, T. Hamada, K. Nishida, S. Tokonami, T. Itoh, H. Miyasaka, and T. Iida, Selective optical assembly of highly uniform nanoparticles by doughnut-shaped beams, Scientific Reports **3**, 10.1038/srep03047 (2013).

[33] K. Michaelian, Microscopic dissipative structuring and proliferation at the origin of life, Heliyon **3**, e00424 (2017).

[34] K. Michaelian and N. S. Padilla, UVC photon-induced denaturing of DNA: A possible dissipative route to archean enzyme-less replication, Heliyon **5**, e01902 (2019).

[35] K. Michaelian, Homochirality through photon-induced denaturing of RNA/DNA at the origin of life, Life **8**, 21 (2018).

[36] K. Michaelian and A. Simeonov, Fundamental molecules of life are pigments which arose and co-evolved as a response to the thermodynamic imperative of dissipating the prevailing solar spectrum, Biogeosciences **12**, 4913 (2015).

[37] P. Wang, L. Robert, J. Pelletier, W. L. Dang, F. Taddei, A. Wright, and S. Jun, Robust growth of escherichia coli, Current Biology **20**, 1099 (2010).

[38] D. Andrieux and P. Gaspard, Nonequilibrium generation of information in copolymerization processes, Proceedings of the National Academy of Sciences **105**, 9516 (2008).

[39] Z. Zeravcic and M. P. Brenner, Self-replicating colloidal clusters, Proceedings of the National Academy of Sciences **111**, 1748 (2014).

[40] P. S. Marcus, S. Pei, C.-H. Jiang, and P. Hassanzadeh, Three-dimensional vortices generated by self-replication in stably stratified rotating shear flows, Physical Review Letters **111**, 10.1103/physrevlett.111.084501 (2013).

[41] J. A. Barranco and P. S. Marcus, Three-dimensional vortices in stratified protoplanetary disks, The Astrophysical Journal **623**, 1157 (2005).

[42] S. S. R. Offner, K. M. Kratter, C. D. Matzner, M. R. Krumholz, and R. I. Klein, THE FORMATION OF LOW-MASS BINARY STAR SYSTEMS VIA TURBULENT FRAGMENTATION, The Astrophysical Journal **725**, 1485 (2010).

[43] P. Bak and K. Sneppen, Punctuated equilibrium and criticality in a simple model of evolution, Physical Review Letters **71**, 4083 (1993).

[44] S. A. Benner, Artificial life II. a proceedings volume in the santa fe institute studies on the sciences of complexity. herausgegeben von c. g. langton. c. taylor, j. d. farmer und s. rasmussen. addison-wesley, redwood city, CA (USA), 1992. XXI, 854 s., geb. 57.95 $. – ISBNO-201–52570-4, Angewandte Chemie **105**, 1433 (1993).

[45] H. Ueno, T. Suzuki, K. Kinosita, and M. Yoshida, ATP-driven stepwise rotation of FoF1-ATP synthase, Proceedings of the National Academy of Sciences **102**, 1333 (2005).

[46] R. Dewar, D. Juretić, and P. Županović, The functional design of the rotary enzyme ATP synthase is consistent with maximum entropy production, Chemical Physics Letters **430**, 177 (2006).

[47] L. Fleming and D. W. McShea, Drosophilamutants suggest a strong drive toward complexity in evolution, Evolution & Development **15**, 53 (2013).